

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0773

Lincoln 100 kV
July 16, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0773. The Interconnection Customer proposes a 55.7 MW battery energy storage facility interconnecting at the Lincoln 100 kV substation in the Illinois area. The request was received on April 08, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 319 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,140,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: NAGUABO 2701 – PEARSON RD 100.0 kV
Contingency: LINCOLN 100 kV/1 kV Transformer Bank
Loading: 105.4% (Rating: 121 MVA)
DFAX: 43.3%
Pre-project loading: 85.4%
2. Facility: OLD FORGE – RAVENNA 230.0 kV
Contingency: RAVENNA – HYLEBOS 500.0 kV
Loading: 103.4% (Rating: 834 MVA)
DFAX: 8.6%
Pre-project loading: 102.8%
3. Facility: ONI-HACKBERRY 500 kV/230 kV Transformer
Contingency: OMNI-HACKBERRY SUBSTATION – NORWIH 500.0 kV
Loading: 103.1% (Rating: 942 MVA)
DFAX: 5.5%
Pre-project loading: 102.8%
4. Facility: NAGUABO 2701 – BRINGHURST 100.0 kV
Contingency: LINCOLN 100 kV/1 kV Auto
Loading: 102.4% (Rating: 105 MVA)
DFAX: 43.3%
Pre-project loading: 79.5%
5. Facility: LINCOLN 100 kV/1 kV Transformer Bank
Contingency: BELRIDGE 1A – LAKE MIRROR 100.0 kV
Loading: 101.3% (Rating: 274 MVA)
DFAX: 59.7%
Pre-project loading: 89.2%

6. Facility: MOORHEAD – E J STONEMAN STA 100.0 kV Ckt 2
Contingency: MOORHEAD – E J STONEMAN STA 100.0 kV Ckt 1
Loading: 100.2% (Rating: 143 MVA)
DFAX: 26.7%
Pre-project loading: 89.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NAGUABO 2701 – PEARSON RD 100.0 kV L	Reconductor line	\$489,000	88% (\$429,000)
OLD FORGE – RAVENNA 230.0 kV	Reconductor line	\$5,543,000	1% (\$33,000)
OMNI-HACKBERRY 500-230 kV Auto	Replace transformer with higher-rated unit	\$24,231,000	1% (\$142,000)
NAGUABO 2701 – BRINGHURST 100.0 kV	Reconductor line	\$1,249,000	88% (\$1,097,000)
LINCOLN 100 kV/1 kV Autotransformer	Replace transformer with higher-rated unit	\$3,140,000	92% (\$2,872,000)
MOORHEAD – E J STONEMAN STA 100.0 kV Ckt 2	Reconductor line	\$694,000	82% (\$567,000)
	Total allocated to project		\$5,140,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.